FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2015**

HALF COURSE TEST –VI

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 09)** contains 09 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (10 – 13) contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (14 – 19) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (1 – 03)** contains 3 Matrix Match Type (4 × 4 Matrix) questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+6 marks** for all correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

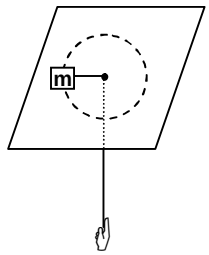
SECTION – A

Single Correct Choice Type

*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. A particle moves along the positive branch of the curve $y = \frac{x^2}{2}$ with x governed by $x = \frac{t^2}{2}$ where x and y are in meter and t in second. At t = 2 the acceleration of the particle is
(A) $\hat{i} - 6\hat{j}$ (B) $\hat{i} + 6\hat{j}$
(C) $2\hat{i} + 4\hat{j}$ (D) $3\hat{i} - 6\hat{j}$

2. A block of mass 'm' is placed on a frictionless horizontal table and attached to a string passing through a small hole in the surface. Initially the mass moves in a circle of radius r_0 with a speed v_0 and the string is held by a person. The person pulls the string slowly to decrease the radius of the circle to r. The tension in the string depends on r as :
(A) r^{-3} (B) r^3
(C) r (D) r^{-1}

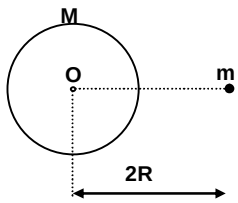


3. Two identical rods are joined to form a 'X'. The smaller angle between the rod is θ. The moment of inertia of the system about an axis passing through the point of intersection of the rods and perpendicular to their plane is proportional to
(A) θ (B) $\sin^2\theta$
(C) $\cos^2\theta$ (D) independent of θ

4. Three identical solid spheres move down on three fixed inclined planes A, B, and C all of the same dimensions A is without friction, the friction between B and sphere is sufficient to cause rolling without slipping, the friction between C and sphere cause rolling with slipping. The kinetic energies of sphere at the bottom of the inclines A, B and C are E_A, E_B and E_C respectively, then
(A) $E_A = E_B = E_C$ (B) $E_A = E_B > E_C$
(C) $E_A > E_B > E_C$ (D) $E_A > E_B = E_C$

Space for rough work

5. A particle of mass m is at a distance 2R from the centre of a thin shell of mass M and having radius R as shown in figure. The gravitational field at the centre of shell is



(A) zero (B) $\frac{GM}{R^2}$

(C) $\frac{G(M+m)}{4R^2}$ (D) $\frac{Gm}{4R^2}$

6. An organ pipe of cross sectional area 100 cm^2 resonates with a tuning fork of frequency 1000 Hz in fundamental tone. The minimum volume of water to be drain out so that the pipe again resonate with the same tuning fork is (take velocity of wave = 320 m/s)
(A) 800 cm^3 (B) 1200 cm^3
(C) 1600 cm^3 (D) 2000 cm^3

7. An elevator car starts descending with constant acceleration 2 m/s^2, 2 second after the start a ball is thrown horizontally with 3 m/s with respect to elevator from point P, then the magnitude of displacement of the ball w.r.t. the point P after 1 sec is
(A) 5 m (B) 10 m
(C) 3 m (D) 6 m

8. The standing wave on a 5 m long string clamped at both ends is represented by the equation y = 4 sin 6.28 x cos 6.28 t, where y is in cm. The phase difference between two points at x = 1.51 m and x = 2.75 m is
(A) π (B) zero
(C) $\pi/2$ (D) $3\pi/2$

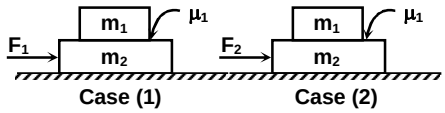
9. An ideal diatomic gas undergoes a process in which the heat absorbed by the gas is twice the increment of its internal energy. Then the polytropic exponent of the process is
(A) 2 (B) 3/5
(C) 2/5 (D) 5/2

Space for rough work

Assertion - Reason Type

This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ***ONLY ONE*** *is correct.*

10. STATEMENT-1: A force F_1 is applied on the lower block in case (1) due to which only lower block moves with constant velocity.



A force F_2 is applied on the lower block in case (2) due to which both the block moves with constant velocity. F_1 and F_2 will be equal. (Given that nature of surfaces is same for both the cases)

because

STATEMENT-2: Frictional force between ground and m_2 will be same for both the case.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

11. STATEMENT-1: When one object collides with another object, the impulse during deformation and restitution will be in same direction.

because

STATEMENT-2: Due to this impulse the objects first deform and due to the same impulse they again try to regain its original shape.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: Rolling without slipping can not be possible in absence of friction.

because

STATEMENT-2: During rolling without slipping, energy can remain conserved in absence of external applied force.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: In absence of air friction, it is claimed that all objects fall with the same acceleration although, a heavier object is pulled towards the earth with more force than a lighter object.

because

STATEMENT-2: Net external force is always equal to rate of change of linear momentum.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

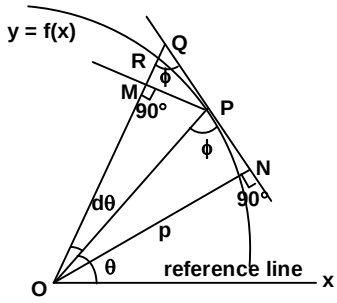
Paragraph for Question Nos. 14 to 16

If a point P moves in plane along a given curve y = f(x), the angular velocity of point P about a fixed point O in the plane is the rate of change of the angle that OP line makes with a fixed direction $[OX-\text{line}]$ in the plane

Let OP = r at t = t sec

PM = $rd\theta$ = PQ $\sin\phi$, But if $d\theta$ is very small then. PQ $\cong \overset{\frown}{PR} = ds$ (arc length)

$\Rightarrow rd\theta = ds \sin\phi$



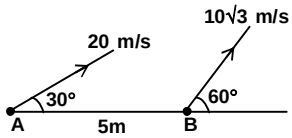
$$\Rightarrow \frac{d\theta}{dt} = \frac{1}{r}\frac{ds}{dt}\sin\phi = \frac{v\sin\phi}{r} = \frac{\text{Magnitude of component of velocity of point perpendicular to radius vector}}{\text{Magnitude of radius vector}}$$

$$\Rightarrow \frac{d\theta}{dt} = \frac{vp}{r^2}$$

Read the above passage carefully and answer the following questions

14. $r^2\frac{d\theta}{dt}$ represents
(A) rate at which radius vector sweeps out area
(B) angular momentum
(C) moment of velocity about origin
(D) rate of increase of sectional area as P moves along curve

15. If two particles A and B are having speed $10\sqrt{3}$ m/s and 20 m/s at a particular instant as shown in the figure, then the angular velocity of A with respect to B at the same instant is



(A) 1 rad/s clockwise
(B) 1 rad/s anticlockwise
(C) 2 rad/s clockwise
(D) 2 rad/s anticlockwise

16. If point P moves on parabolic path $y^2 = 4(x + 1)$, where x and y are in meter with constant speed 2 m/s. Its angular velocity about focus at an instant when it makes angle 60° at focus with x-axis is [all angles are measured in anticlockwise direction with positive x-axis]
(A) 0.25 rad/s
(B) 0.50 rad/s
(C) 0.12 rad/s
(D) none of the above

Space for rough work

Paragraph for Question Nos. 17 to 19

A cubical box of side 2 metre contains oxygen gas (atomic weight 32) at a pressure of 100N/m^2. During an observation time of one second, an atom travelling with the root-mean square speed parallel to one of the edges of the cube was found to make 500 hits with a particular wall, without any collision with the other atoms. ($R = \frac{25}{3}$ J/mol–K and $K = 1.38 \times 10^{-23}$ J/K)

17. What is the temperature of the gas
(A) 500 K (B) 4520 K
(C) 5120 K (D) 3600 K

18. What is the average translational kinetic energy per atom
(A) 106×10^{-21} J (B) 96×10^{-21} J
(C) 116×10^{-21} J (D) 112×10^{-21} J

19. Find the total mass of oxygen gas in the box.
(A) 30×10^{-4} kg (B) 2.4×10^{-4} kg
(C) 6×10^{-4} kg (D) 15×10^{-4} kg

Space for rough work

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A particle is moving according to the displacement time relation $x = 3t^2 - \frac{t^3}{2}$ (where x is in meters and t is in seconds). Match the condition of column I with time interval and instant of column II

Column A		Column B	
(A)	Velocity and acceleration will be in same direction	(p)	At t = 0 and t = 6 sec
(B)	particle will be at origin	(q)	0 < t < 2 sec
(C)	particle will retard	(r)	at t = 0 and t = 4 sec
(D)	velocity is zero	(s)	2 < t < 4

Space for rough work

2. A bob is attached to a string of length ℓ whose other end is fixed and is given horizontal velocity at the lowest point of the circle so that the bob moves in a vertical plane. Match the velocity given at the lowest point of circle in column I with tension and velocity at the highest point of the circle corresponding to velocity of column I of column II

Column A	Column B
(A) $\sqrt{2g\ell}$	(p) $T = mg,\ v > \sqrt{g\ell}$
(B) $\sqrt{6g\ell}$	(q) $T > 2mg$ and $v > \sqrt{3g\ell}$
(C) $\sqrt{8g\ell}$	(r) $T = 0$ and $v = \sqrt{g\ell}$
(D) $\sqrt{5g\ell}$	(s) $T = 0$

3. Match the following.

Column – A	Column – B
(A) Ideal gas	(p) Maximum emissivity
(B) Black body	(q) Maximum absorptive power
(C) Wien's displacement law	(r) An ideal concept
(D) A red object having 0 K temperature.	(s) Dominant wave in composite radiations.

Space for rough work

Chemistry

PART - II

SECTION – A

Single Correct Choice Type

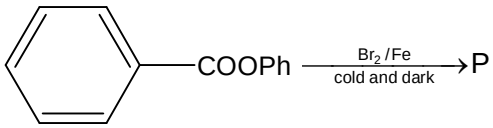
*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. $Ag(CN)_2^-$ is stable while $AgCl_2^-$ is unstable because
   (A) CN^- is stronger than Cl^-
   (B) Ag^+ is soft acid, CN^- is soft base while Cl^- is hard base
   (C) both are equally stable
   (D) None of the above

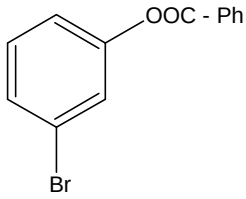
2. Glycerol on oxidation with Fenton's reagent produces
   (A) glyceraldehyde
   (B) dihydroxyacetone
   (C) tartaric acid
   (D) glyceraldehyde and dihydroxyacetone

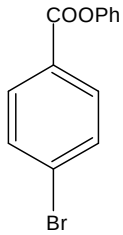
3. Consider following reaction,
   $Zn + 2H^+ \longrightarrow Zn^{2+} + H_2$
   half-life period is independent of concentration of Zn at constant pH. At constant Zn concentration, half-life is 10 minutes at pH = 2 and half-life is 100 minutes at pH = 3. Hence, rate law is
   (A) $k\,[Zn]\,[H^+]$
   (B) $k\,[Zn]\,[H^+]^2$
   (C) $k\,[Zn]^0\,[H^+]$
   (D) $k\,[Zn]^0\,[H^+]^2$

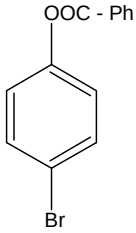
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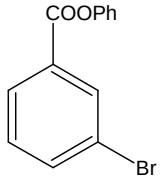
4. 

(a monobromo derivative), where P is

(A) 

(B) 

(C) 

(D) 

5. Which of the following acid contains one S – S bond?
(A) $H_2S_2O_7$ (B) $H_2S_2O_8$
(C) $H_2S_2O_6$ (D) none of these

6. Which of the following plots represent the behaviour of an ideal binary liquid solution?
(A) Plot of P_{Total} vs Y_A is linear (B) Plot of $\frac{1}{P_{Total}}$ vs Y_B is linear
(C) Plot of P_{Total} vs Y_B is linear (D) None of the above

7. The value of $\frac{RT}{PV}$ for a gas at critical condition is how many times of $\frac{RT}{PV}$ at normal condition?
(A) $\frac{8}{3}$ (B) $\frac{3}{8}$
(C) 3 (D) $\frac{1}{3}$

Space for rough work

8. In a closed vessel, ozone transforms into oxygen through two steps as mentioned below:

$O_3 \rightleftharpoons O_2 + O$ (fast)

$O + O_3 \longrightarrow 2O_2$ (slow)

If half of the oxygen produced is removed from the vessel without disturbing the temperature, then the rate of transformation of ozone into oxygen will be

(A) unaffected
(B) half the initial value
(C) double the initial rate
(D) None of the above

9. The stoichiometric equation for the oxidation of bromide ions by hydrogen peroxide in acidic solution is

$2Br^- + H_2O_2 + 2H^+ \longrightarrow Br_2 + 2H_2O$

$r = k\,[H_2O_2]\,[H^+]\,[Br^-]$

If, by the addition of water, the total volume were doubled, what would be the effect on the rate of disappearance of Br^- and rate of reaction?

(A) Rate of disappearance of Br^- becomes $\frac{1}{8}$ times and rate of reaction $\frac{1}{4}$ times

(B) Rate of disappearance of Br^- becomes $\frac{1}{4}$ times and rate of reaction $\frac{1}{8}$ times

(C) Both rate of disappearance of Br^- and rate of reaction becomes $\frac{1}{8}$ times

(D) Rate of disappearance of Br^- becomes 4 times and rate of reaction 8 times

Space for rough work

Assertion - Reason Type

*This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

10. STATEMENT-1: The pH of pure water is less than 7 at 60^oC
and
STATEMENT-2: As the temperature increases, pure water becomes slightly acidic
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

11. STATEMENT-1: CO_2 is a gas and SiO_2 is a solid at room temperature.
and
STATEMENT-2: van der Waals' forces are far weaker than ordinary covalent bond.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: Acidic strength boron halide is as $BF_3 < BCl_3 < BBr_3 < BI_3$
and
STATEMENT-2: Fluorine is more electronegative & creates greater electron deficiency on boron.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

13. STATEMENT-1: R.M.S velocity of a gas sample can be expressed as $C_{rms} = \sqrt{\frac{3PV}{M}}$
and
STATEMENT-2: The R.M.S velocity of a gas sample decreases if its volume is decreased.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

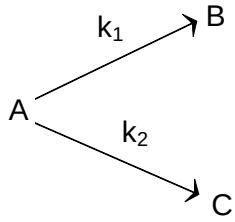
Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 14 to 16

Parallel or competing reactions–

The reactions in which a substance reacts or decomposes in more than one manner are called parallel or side reactions. This type of kinetics can be observed in radioactive conversions where an element loses α, β particles simultaneously to form new products at the same time through different paths.



Then, $-\frac{d[A]}{dt} = (K_1 + K_2)\,[A] = K_{av.}\,[A]$

K_1 = Fractional yield of B × $K_{av.}$

K_2 = Fractional yield of C × $K_{av.}$

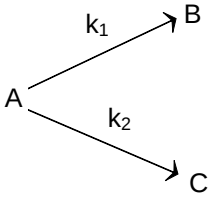
If $K_1 >> K_2$, then

$A \rightarrow B$ will become main reaction

$A \rightarrow C$ will become side reaction

Read the above paragraph carefully and answer the questions given below it:

14. A substance undergoes first order decomposition. The decomposition follows two parallel first order reactions as–



$K_1 = 1.26 \times 10^{-4}\ \text{sec}^{-1}$

$K_2 = 3.8 \times 10^{-5}\ \text{sec}^{-1}$

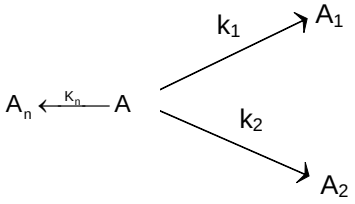
The percentage distribution of B and C are–

(A) 75% B and 25% C (B) 80% B and 20% C

(C) 76.83% B and 23.17% C (D) 90% B and 10% C

Space for rough work

15. For a first order decomposition–



Overall K will be given by–

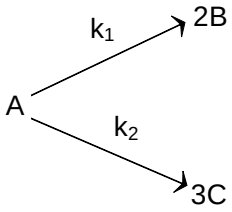
(A) $K = K_1 + K_2 + \ldots\ldots\ldots + K_n$

(B) $K = \dfrac{(K_1 + K_2 + \ldots\ldots + K_n)}{n}$

(C) $K = K_1 \times K_2 \times K_3 \ldots\ldots\ldots\ldots \times K_n$

(D) None

16. For a reaction,



Follow parallel kinetics, the ratio of [B] and [C] at any instant can be given by,

(A) $\dfrac{K_1}{K_2}$

(B) $\dfrac{K_2}{K_1}$

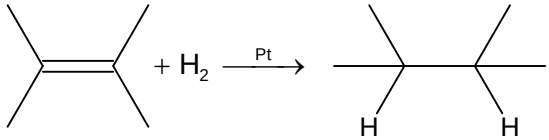
(C) $\dfrac{2K_1}{3K_2}$

(D) $\dfrac{3K_1}{2K_2}$

Space for rough work

Paragraph for Question Nos. 17 to 19

The *cis* and *trans* isomers of alkene do not have the same stability. The stability can be measured by hydrogenation and combustion. The reaction of alkene with hydrogen is endothermic and the enthalpy change in the reaction is called heat of hydrogenation.

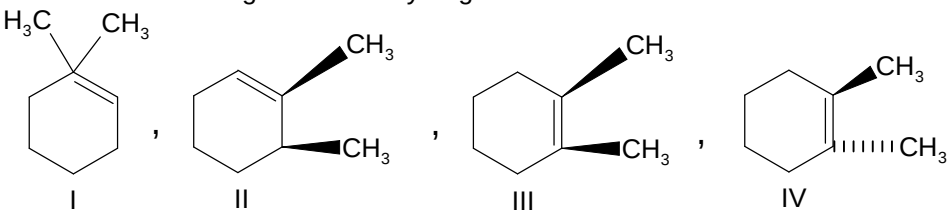


$\Delta H = -120\,kJ/mol$

In all the isomers of 2–butene, the product is same but different amount of heat is evolved. In each reaction, it must be related with different relative stability. 1–butene evolves greatest amount of energy and *trans* 2–butene evolves least amount of energy. So, 1–butene must have greatest energy and it is least stable whereas trans 2–butene must have lowest energy and it is more stable. *cis* 2–butene has intermediate energy in relation to above.

Read the above paragraph carefully and answer the questions given below:

17. Which alkene has higher heat of hydrogenation?

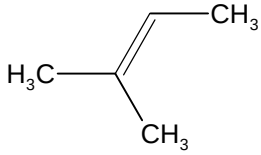


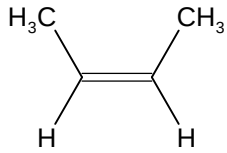
I , II , III , IV

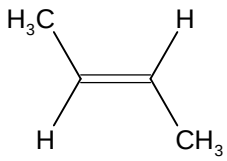
(A) I (B) II
(C) III (D) IV

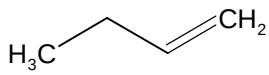
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18. Which alkene is most stable?

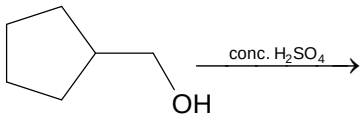
(A) 

(B) 

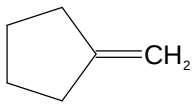
(C) 

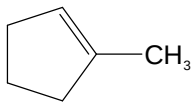
(D) 

19. For the reaction,

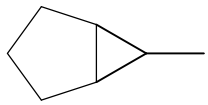


the major product is

(A) 

(B) 

(C) 

(D) 

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SECTION – B
(Matching List Type)

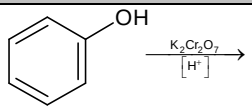
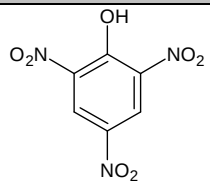
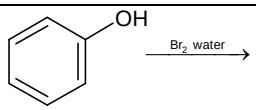
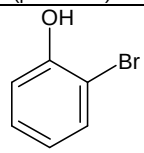
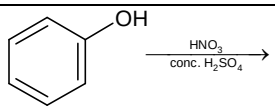
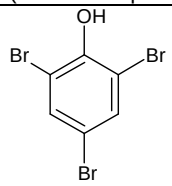
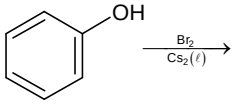
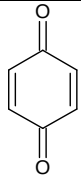
This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Match Column–I and Column –II.

Column–I		Column–II	
(A)	The volume of mixture has no effect on the equilibrium constant.	(p)	Common ion effect
(B)	Increasing the pressure moves the equilibrium to the left.	(q)	pH = 4
(C)	The solution has hydronium ion concentration of 0.0001 mol/litre.	(r)	There is an increase in the number of moles from products to reactants.
(D)	The addition of NaOH to $Ca(OH)_2$ solution precipitates $Ca(OH)_2$.	(s)	There is no change in number of moles.

Space for rough work

2. Match the following:

Column – I		Column – II	
(A)		(p)	 (picric acid)
(B)		(q)	 (ortho-bromophenol)
(C)		(r)	 (tribromophenol)
(D)		(s)	 (para-benzoquinone)

3. Match the following:

Column – I (Thermo chemical equations)		Column – II (Characteristics)	
(A)	$CuSO_4.5H_2O + \text{Excess } H_2O + 2.9 \text{ kcal} \longrightarrow CuSO_4.5H_2O(aq)$	(p)	Enthalpy change $\Delta H < 0$
(B)	$2H_2(g) + O_2(g) \longrightarrow 2H_2O(\ell) + 115.0 \text{ kcal}$	(q)	Enthalpy of combustion
(C)	$H_2SO_4(aq) + NaOH(aq) \longrightarrow Na_2SO_4(aq) + H_2O + 57.1 \text{ kJ}$	(r)	Enthalpy of solvation
(D)	$CH_4(g) + 2O_2(g) \longrightarrow 2H_2O(g) + CO_2(g) + 803.3 \text{ kJ}$	(s)	Enthalpy of neutralization

Space for rough work

Mathematics PART – III

SECTION – A

Single Correct Choice Type

*This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. Maximum sum of coefficient in the expansion of $(1 - x \sin\theta + x^2)^n$ is
   (A) 1 (B) 2^n
   (C) 3^n (D) 0

2. The number of integral solutions of $x + y + z = 0$ with $x \geq -5$, $y \geq -5$, $z \geq -5$ is
   (A) 134 (B) 136
   (C) 138 (D) 140

3. Double ordinate AB of the parabola $y^2 = 4ax$ subtends an angle $\frac{\pi}{2}$ at the focus of the parabola, then tangents drawn to parabola at A and B will intersect at
   (A) (–4a, 0) (B) (–2a, 0)
   (C) (–3a, 0) (D) none of these

4. The tangent at a point P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets one of the directrix in F. If PF subtends an angle θ at the corresponding focus, then θ equals
   (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
   (C) $\frac{3\pi}{4}$ (D) π

Space for rough work

5. If in triangle ABC, A ≡ (1, 10), circumcentre ≡ $\left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocentre ≡ $\left(\frac{11}{4}, \frac{4}{3}\right)$, then the coordinates of mid–point of side opposite to A is

(A) $\left(1, -\frac{11}{3}\right)$ (B) (1, 5)

(C) (1, –3) (D) (1, 6)

6. $\cos^3 x \sin 2x = \sum_{x=0}^{n} a_r \sin(rx) \ \forall\ x \in R$, then

(A) n = 5, $a_1 = \frac{1}{2}$ (B) n = 5, $a_1 = \frac{1}{4}$

(C) n = 5, $a_2 = \frac{1}{8}$ (D) n = 5, $a_2 = \frac{1}{4}$

7. If the roots of the quadratic equation $(4p - p^2 - 5)x^2 - (2p - 1)x + 3p = 0$ lie on either side of unity, then the number of integral values of p is

(A) 1 (B) 2

(C) 3 (D) 4

8. If $z = (\lambda + 3) - i\sqrt{5 - \lambda^2}$, then the locus of z is

(A) ellipse (B) semicircle

(C) parabola (D) straight line

9. If $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + ... = \frac{\pi}{4}$, then value of $\frac{1}{1\times 3} + \frac{1}{5\times 7} + \frac{1}{9\times 11} + ...$ is

(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{36}$

Space for rough work

Assertion - Reason Type

*This section contains 4 questions numbered 10 to 13. Each question contains STATEMENT – 1 (Assertion) and STATEMENT -2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

10. STATEMENT 1: Principal value of $\cos^{-1}(\cos 30)$ is $30 - 9\pi$
STATEMENT 2: $30 - 9\pi \in [0, \pi]$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

11. STATEMENT 1: If there is exactly one point on the line $3x + 4y + 5\sqrt{5} = 0$, from which perpendicular tangents can be drawn to the ellipse $\frac{x^2}{a^2} + y^2 = 1$ (a > 1), then the eccentricity of the ellipse is $\frac{1}{3}$
STATEMENT 2: For the condition given in statement 1, given line must touch the circle $x^2 + y^2 = a^2 + 1$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

12. STATEMENT 1: If the point $(2a - 5, a^2)$ is on the same side of the line $x + y - 3 = 0$ as that of the origin, then $a \in (2, 4)$
STATEMENT 2: The points (x_1, y_1) and (x_2, y_2) lie on the same or opposite sides of the line $ax + by + c = 0$, as $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have the same or opposite signs.
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

13. STATEMENT 1: If $px^2 + qx + r = 0$ is a quadratic equation ($p, q, r \in R$) such that its roots are α, β and $p + q + r < 0$, $p - q + r < 0$ and $r > 0$, then $[\alpha] + [\beta] = -1$, (where [.] denotes the greatest integer function)
STATEMENT 2: If for any two real numbers a and b, function f(x) is such that $f(a)\, f(b) < 0$,
$\Rightarrow$ f(x) has at least one real root lying in (a, b)
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Space for rough work

Linked Comprehension Type

*This section contains 2 paragraphs. Based upon each paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
Two consecutive numbers from 1, 2, 3, ..., n are removed. The arithmetic mean of the remaining numbers is $\frac{105}{4}$

14. The value of n lies in
(A) [45, 55] (B) [52, 60]
(C) [41, 49] (D) none of these

15. The removed numbers
(A) lie between 10 and 20 (B) are greater than 10
(C) are less than 15 (D) none of these

16. Sum of all numbers
(A) exceeds 1600 (B) is less than 1500
(C) lies between 1300 and 1500 (D) none of these

Paragraph for Question Nos. 17 to 19

Read the following write up carefully and answer the following questions:
P is a set containing n elements. A subset A of P is chosen and the set P is reconstructed by replacing the elements of A. A subset B of P is chosen again

17. The number of ways of choosing A and B such that A and B have no common elements is
(A) 3^n (B) 2^n
(C) 4^n (D) none of these

18. The number of ways of choosing A and B such that B contains just one element more than A is
(A) 2^n (B) ${}^{2n}C_{n-1}$
(C) ${}^{2n}C_n$ (D) $(3^n)^2$

19. The number of ways of choosing A and B such that B is a subset of A is
(A) ${}^{2n}C_n$ (B) 4^n
(C) 3^n (D) none of these

Space for rough work

SECTION – B
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. If e_1 and e_2 are the roots of the equation $x^2 - ax + 2 = 0$, then match the following Column–I with Column–II

	Column – I		Column – II
(A)	If e_1 and e_2 are the eccentricities of the ellipse, and hyperbola, respectively then the values of 'a' are	(p)	6
(B)	If both e_1 and e_2 are the eccentricities of the hyperbolas, then values of a are	(q)	$\frac{5}{2}$
(C)	If e_1 and e_2 are eccentricities of hyperbola and conjugate hyperbola, then values of a are	(r)	$2\sqrt{2}$
(D)	If e_1 is the eccentricity of the hyperbola for which there exist infinite points from which perpendicular tangents can be drawn and e_2 is the eccentricity of the hyperbola in which no such points exist then the values of a are	(s)	5

2. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	$\left(\sin^{-1}x\right)^2 + \left(\sin^{-1}y\right)^2 = \frac{\pi^2}{2} \Rightarrow x^3 + y^3$ is equal to	(p)	1
(B)	$\left(\cos^{-1}x\right)^2 + \left(\cos^{-1}y\right)^2 = 2\pi^2 \Rightarrow x^5 + y^5$ is equal to	(q)	–2
(C)	$\left(\sin^{-1}x\right)^2\left(\cos^{-1}y\right)^2 = \frac{\pi^4}{4} \Rightarrow \lvert x - y\rvert$ is equal to	(r)	0
(D)	$\left\lvert\sin^{-1}x - \sin^{-1}y\right\rvert = \pi \Rightarrow x^y$ is equal to	(s)	2

3. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	$z^4 - 1 = 0$	(p)	$z = \cos\frac{\pi}{8} + i\sin\frac{\pi}{8}$
(B)	$z^4 + 1 = 0$	(q)	$z = \cos\frac{\pi}{8} - i\sin\frac{\pi}{8}$
(C)	$iz^4 + 1 = 0$	(r)	$z = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$
(D)	$iz^4 - 1 = 0$	(s)	$z = \cos 0 + i \sin 0$

Space for rough work